

Lecture 29/Module 3

Sinusoidal Steady State Analysis

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Problem 1

$$v(t) = 120 \cos(377t + 45^\circ) \text{ V} \quad \text{and} \quad i(t) = 10 \cos(377t - 10^\circ) \text{ A}$$

find the instantaneous power and the average power absorbed by the passive linear network of Fig. 11.1.

Solution:

The instantaneous power is given by

$$p = vi = 1200 \cos(377t + 45^\circ) \cos(377t - 10^\circ)$$

Applying the trigonometric identity

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

gives

$$p = 600 [\cos(754t + 35^\circ) + \cos 55^\circ]$$

or

$$p(t) = 344.2 + 600 \cos(754t + 35^\circ) \text{ W}$$

The average power is

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} 120(10) \cos[45^\circ - (-10^\circ)] \\ &= 600 \cos 55^\circ = 344.2 \text{ W} \end{aligned}$$

which is the constant part of $p(t)$ above.

Problem 2

Calculate the average power absorbed by an impedance $\mathbf{Z} = 30 - j70 \, \Omega$ when a voltage $\mathbf{V} = 120 \angle 0^\circ$ is applied across it.

Solution:

The current through the impedance is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{120 \angle 0^\circ}{30 - j70} = \frac{120 \angle 0^\circ}{76.16 \angle -66.8^\circ} = 1.576 \angle 66.8^\circ \text{ A}$$

The average power is

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} (120)(1.576) \cos(0 - 66.8^\circ) = 37.24 \text{ W}$$

Problem 3

For the circuit shown in Fig. 11.3, find the average power supplied by the source and the average power absorbed by the resistor.

Solution:

The current $\mathbf{I}$ is given by

$$\mathbf{I} = \frac{5\angle 30^\circ}{4 - j2} = \frac{5\angle 30^\circ}{4.472\angle -26.57^\circ} = 1.118\angle 56.57^\circ \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2}(5)(1.118) \cos(30^\circ - 56.57^\circ) = 2.5 \text{ W}$$

The current through the resistor is

$$\mathbf{I}_R = \mathbf{I} = 1.118\angle 56.57^\circ \text{ A}$$

and the voltage across it is

$$\mathbf{V}_R = 4\mathbf{I}_R = 4.472\angle 56.57^\circ \text{ V}$$

The average power absorbed by the resistor is

$$P = \frac{1}{2}(4.472)(1.118) = 2.5 \text{ W}$$

which is the same as the average power supplied. Zero average power is absorbed by the capacitor.

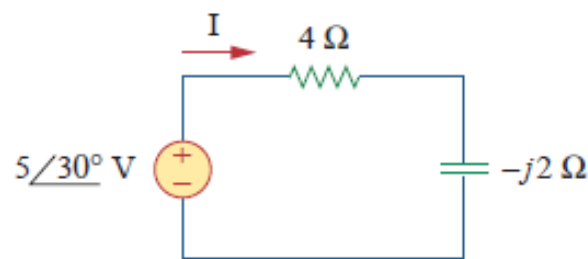


Figure 11.3
For Example 11.3.

Problem 4

A series-connected load draws a current $i(t) = 4 \cos(100\pi t + 10^\circ)$ A when the applied voltage is $v(t) = 120 \cos(100\pi t - 20^\circ)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Solution:

The apparent power is

$$S = V_{\text{rms}} I_{\text{rms}} = \frac{120}{\sqrt{2}} \frac{4}{\sqrt{2}} = 240 \text{ VA}$$

The power factor is

$$\text{pf} = \cos(\theta_v - \theta_i) = \cos(-20^\circ - 10^\circ) = 0.866 \quad (\text{leading})$$

The pf is leading because the current leads the voltage. The pf may also be obtained from the load impedance.

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{120 \angle -20^\circ}{4 \angle 10^\circ} = 30 \angle -30^\circ = 25.98 - j15 \, \Omega$$

$$\text{pf} = \cos(-30^\circ) = 0.866 \quad (\text{leading})$$

The load impedance $\mathbf{Z}$ can be modeled by a 25.98- Ω resistor in series with a capacitor with

$$X_C = -15 = -\frac{1}{\omega C}$$

or

$$C = \frac{1}{15\omega} = \frac{1}{15 \times 100\pi} = 212.2 \, \mu\text{F}$$

Thank You